BITS Pilani

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CE F231
Fluid Mechanics

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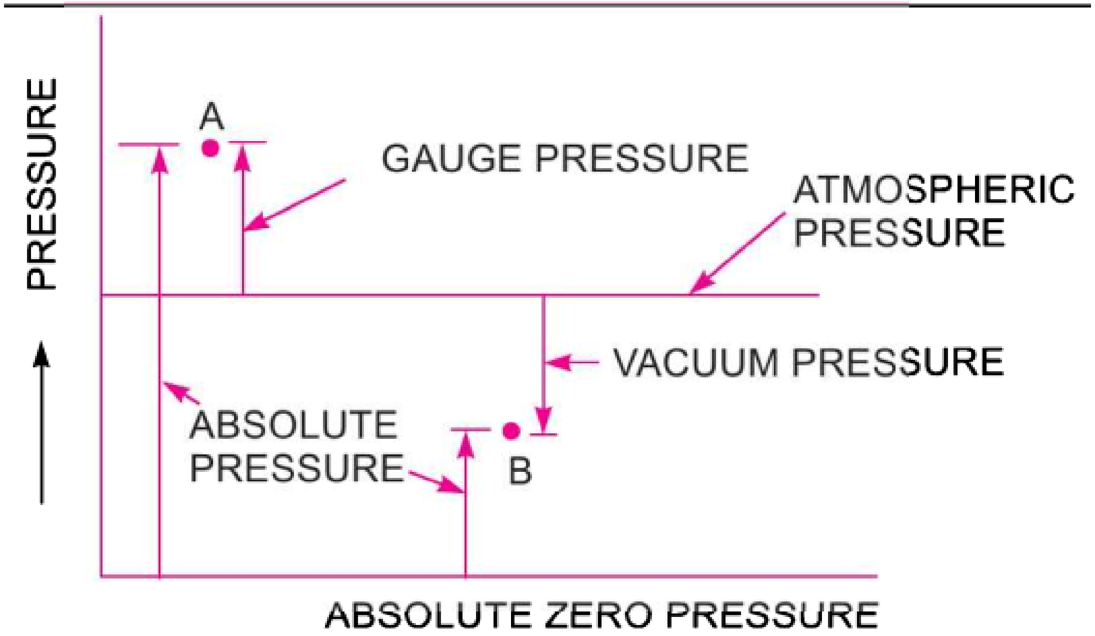
Lecture 5
Pressure and its Measurement

Recap: Absolute, Gauge, Atmospheric, and Vacuum Pressures

innovate achieve lead



Atmospheric pressure 1 atm. = 1.013 bar = 101.3 kN/m^2

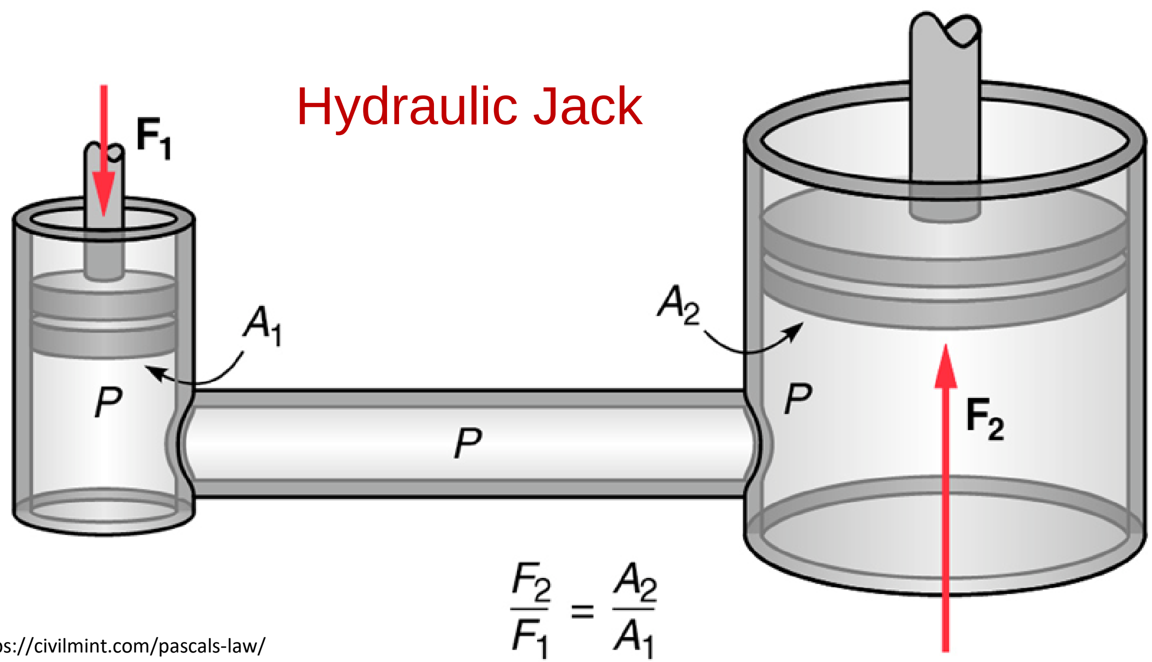
Gauge pressure = Absolute pressure – Atmospheric pressure

Vacuum pressure = Atmospheric pressure – Absolute pressure

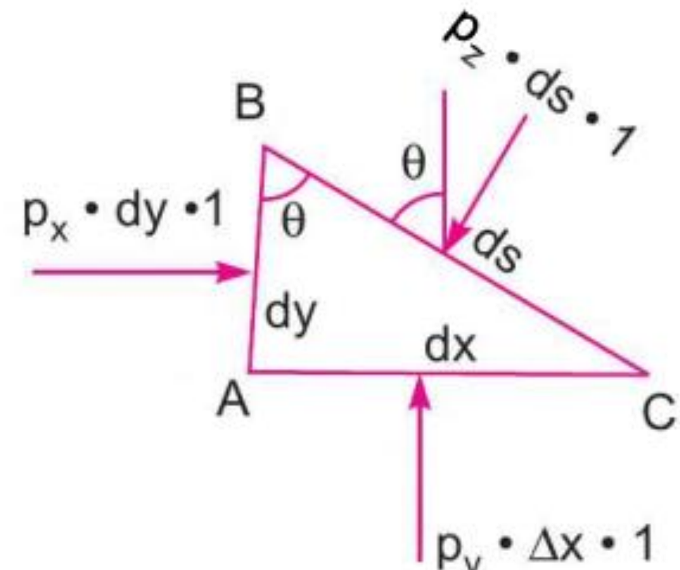
Bansal, R.K., 2005. *A textbook of fluid mechanics and hydraulic machines:(in SI units)*. LAXMI Publications, Ltd..

Recap: Pascal's Law

- Pascal's law states that pressure applied to an enclosed fluid will be transmitted without a change in magnitude to every point of the fluid and to the walls of the container, and the pressure at any point in the fluid is equal in all directions.



https://civilmint.com/pascals-law/



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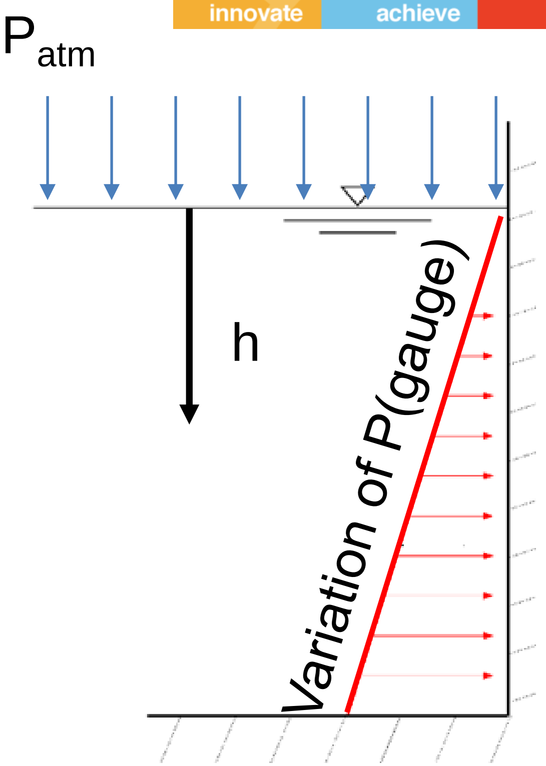
Recap: Hydrostatic Law

- Hydrostatic law states that for a fluid at rest, the rate of increase in pressure in vertically downward direction is equal to the unit weight of the liquid at that point.
- Applicable when gravitational force is the only external body force

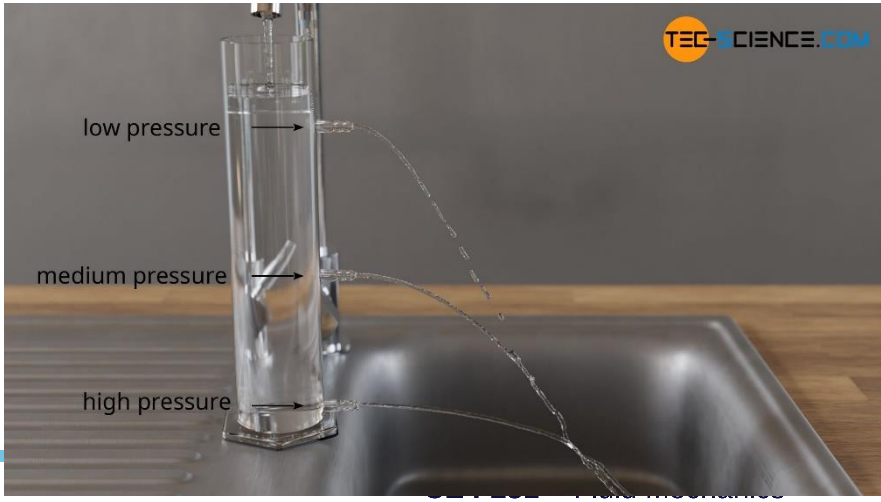
Recap: Hydrostatic Pressure

- $P = h \rho g$
- h = pressure head = $P/(\rho g)$
  - At a depth of 1m below the water surface, what are the gauge and absolute pressures?



https://www.engineeringtoolbox.com/thrust-submerged-surface-d_1767.html

  - If absolute water pressure at a certain depth below free surface is 300 kPa, what is the equivalent head of water?

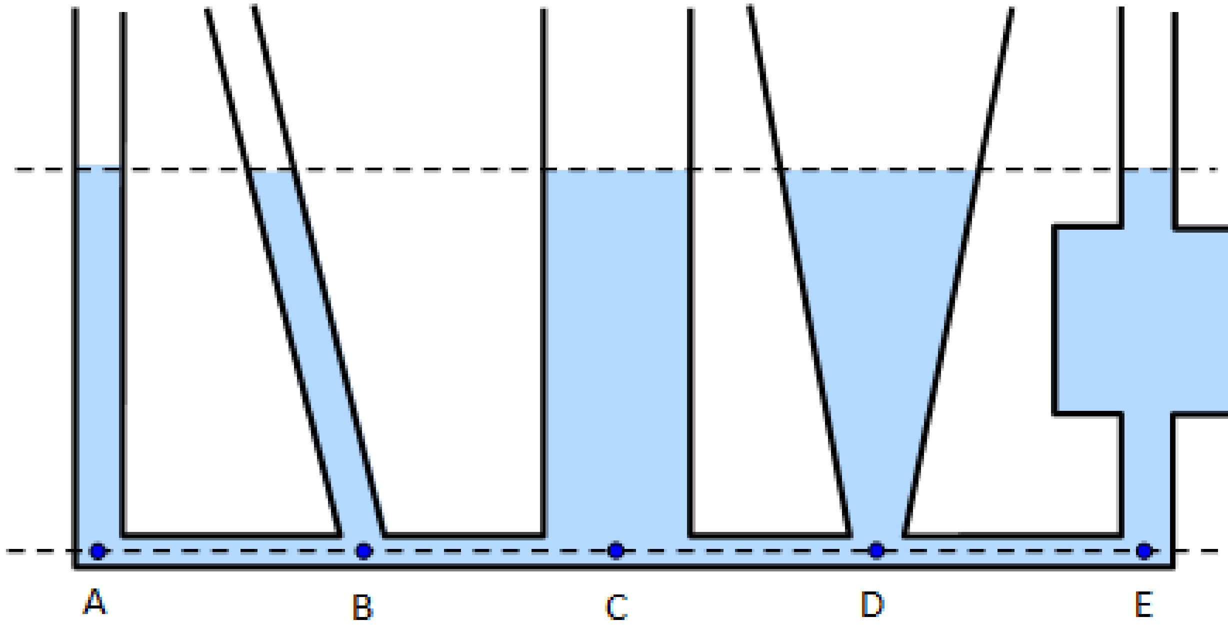


https://www.tec-science.com/mechanics/gases-and-liquids/pressure-in-liquids-hydrostatic-pressure/

Hydrostatic Pressure

- Pressure at a point in a liquid at rest depends on vertical height of liquid above that point

- Pressure in a liquid at rest is same at all points lying on a horizontal line (parallel to free surface of liquid)



https://www.schoolphysics.co.uk/age11-14/Mechanics/Statics/text/Pressure_in_liquids/index.html

Pressure Measurement Using Manometers

- Simple manometers → glass tube with one end connected to pressure measuring point and other end open to atmosphere
  - Piezometers
  - U-tube manometers
  - Single column manometers
    - Vertical single column
    - Inclined single column

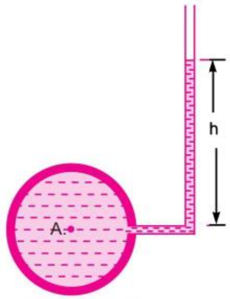
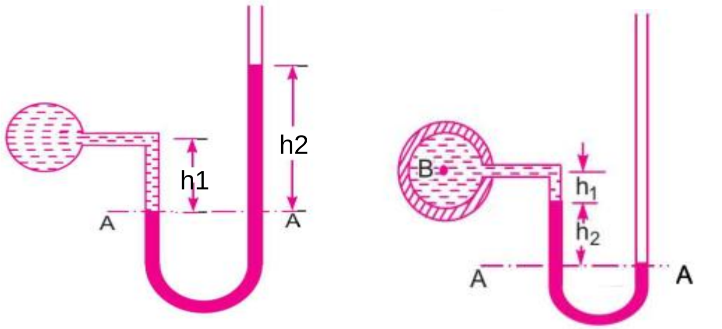


Fig. 2.8 *Piezometer.*



(a) For gauge pressure (b) For vacuum pressure

Fig. 2.9 *U-tube Manometer.*

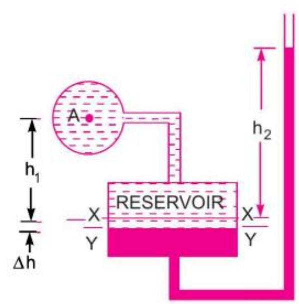


Fig. 2.15 *Vertical single column manometer.*

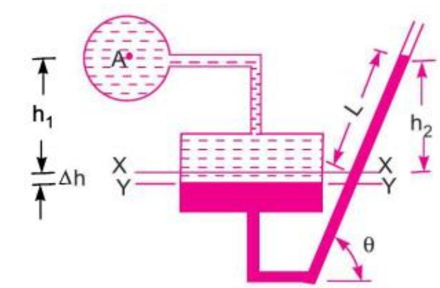


Fig. 2.16 *Inclined single column manometer.*

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Pressure Measurement Using Manometers

- Simple manometers
  - Piezometers
    - Simplest type of manometer for measuring liquid pressure
    - Cannot be used to measure pressures below atmospheric pressure
    - Cannot be used to measure very high pressures
    - Cannot be used to measure gas pressure

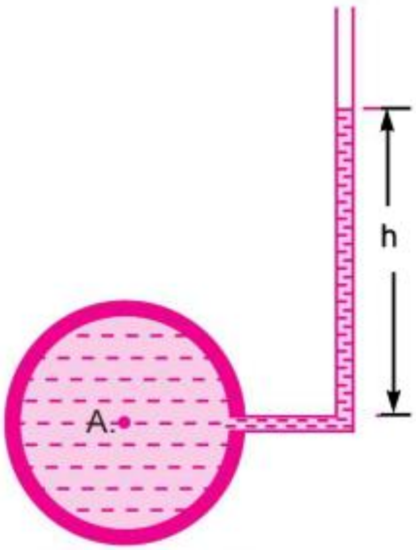

Fig. 2.8 *Piezometer.*

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Pressure Measurement Using Manometers

- Simple manometers
  - U-tube manometers
    - One end connected to pressure measurement point; another end open to atmosphere
    - Fluid used → manometric fluid (e.g., mercury)
    - Measure levels in two arms → estimate pressure at point of interest

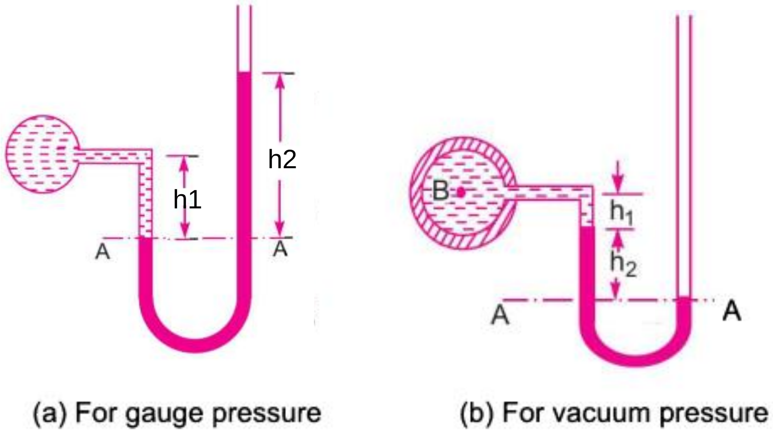
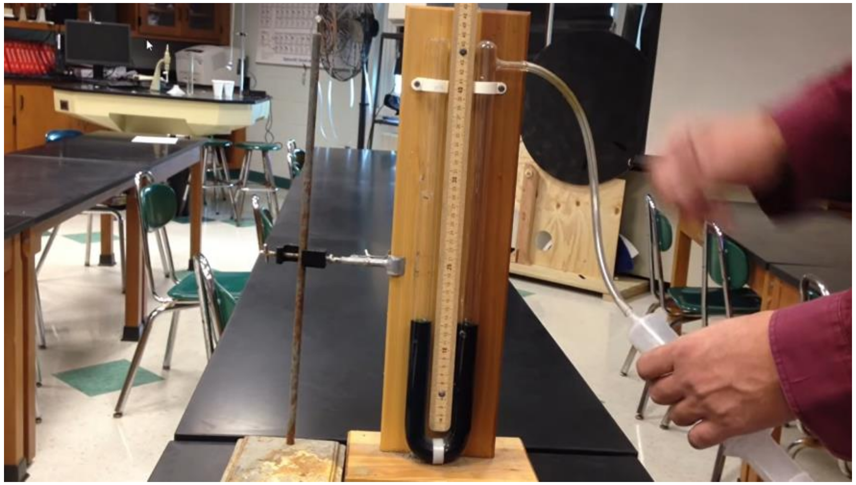


Fig. 2.9 *U-tube Manometer.*



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https://www.youtube.com/watch?v=9SQ2FPHCZJk&t=84s

Pressure Measurement Using Manometers

- Simple manometers
  - Single column manometer
    - Modified U-tube manometer
    - Reservoir cross section area >> limb cross section area (>100 times)
    - Negligible change in elevation in reservoir as compared to other limb
    - Pressure → height of liquid in limb
    - Inclined → increased sensitivity

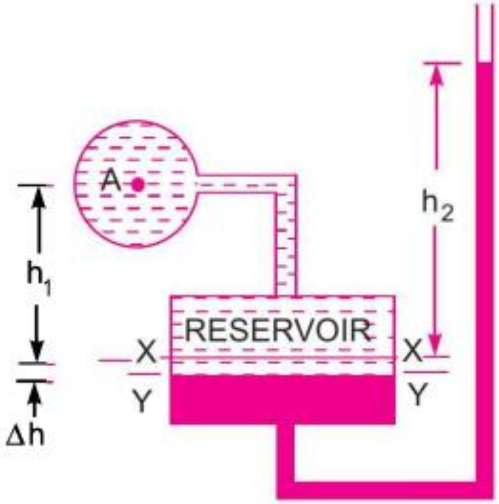

Fig. 2.15 *Vertical single column manometer.*

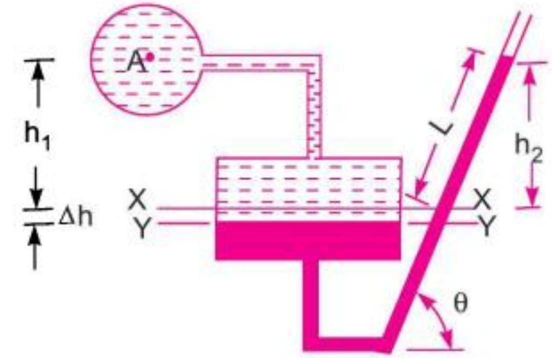

Fig. 2.16 *Inclined single column manometer.*

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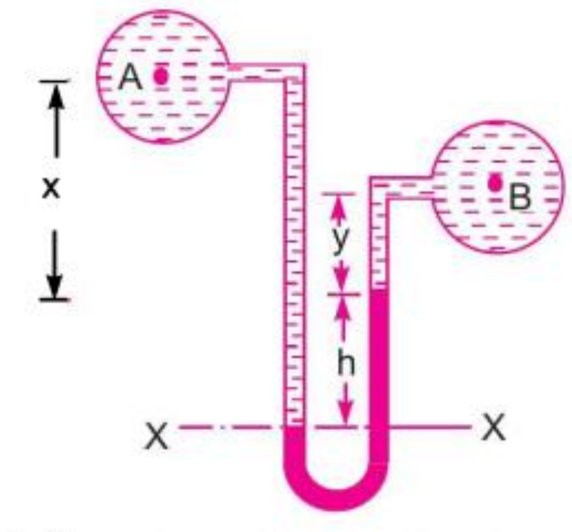
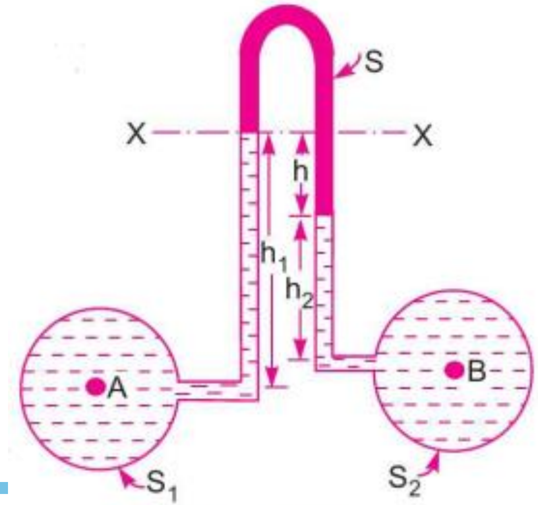
Pressure Measurement Using Manometers

- Differential manometers
  - U-tube differential manometer
    - Sp. gr. of manometric fluid > Sp. gr. of fluids whose pressure difference needs to measured
    - High pressure difference
  - Inverted U-tube differential manometer
    - Sp. gr. of manometric fluid < Sp. gr. of fluids whose pressure difference needs to measured
    - Low pressure difference





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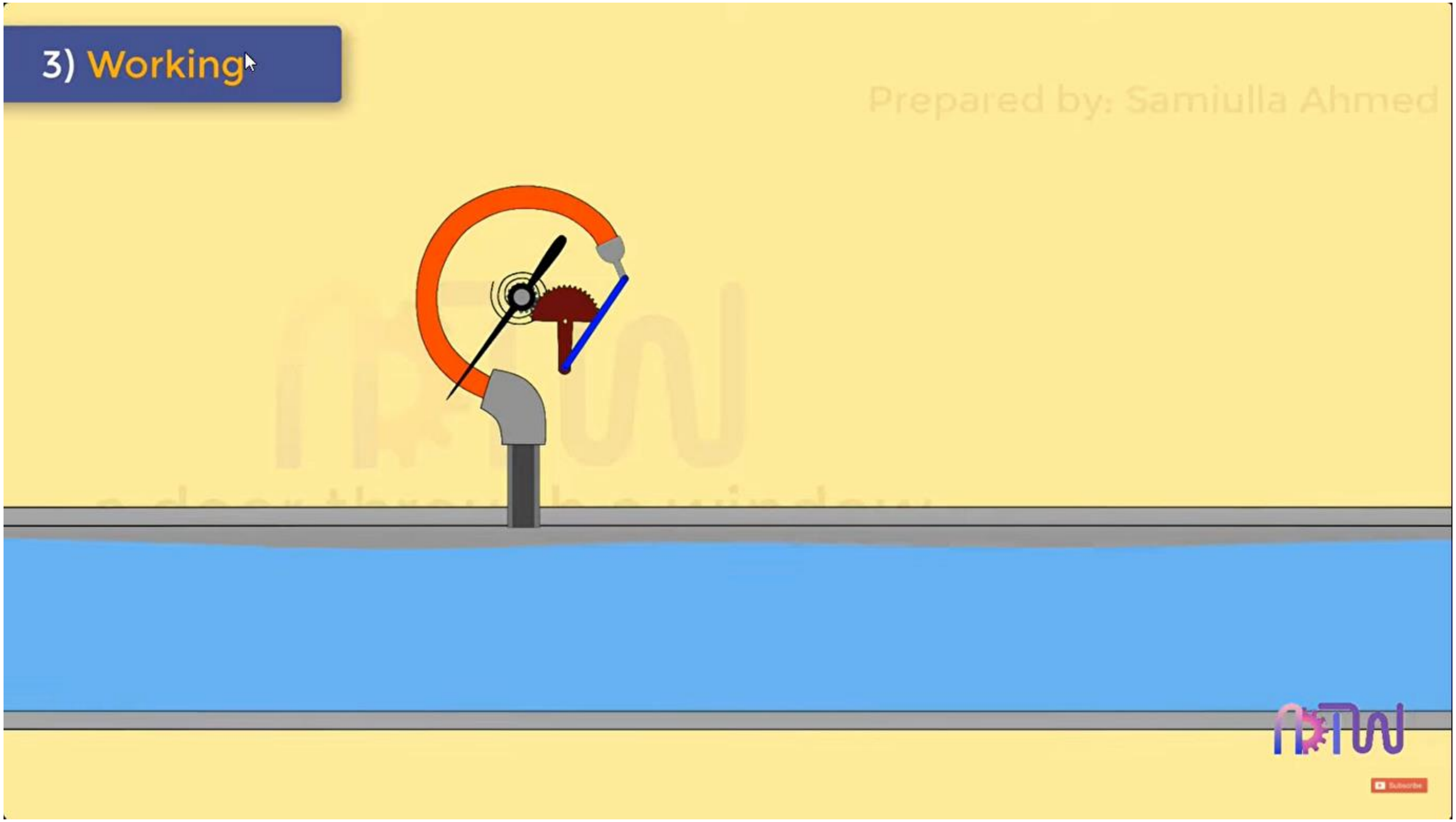
Mechanical Gauges

Bourdon tube pressure gauge (measuring spans 0-0.6 bar to 0-6000 bar)



https://www.youtube.com/watch?v=fmOnrEZ_z6k

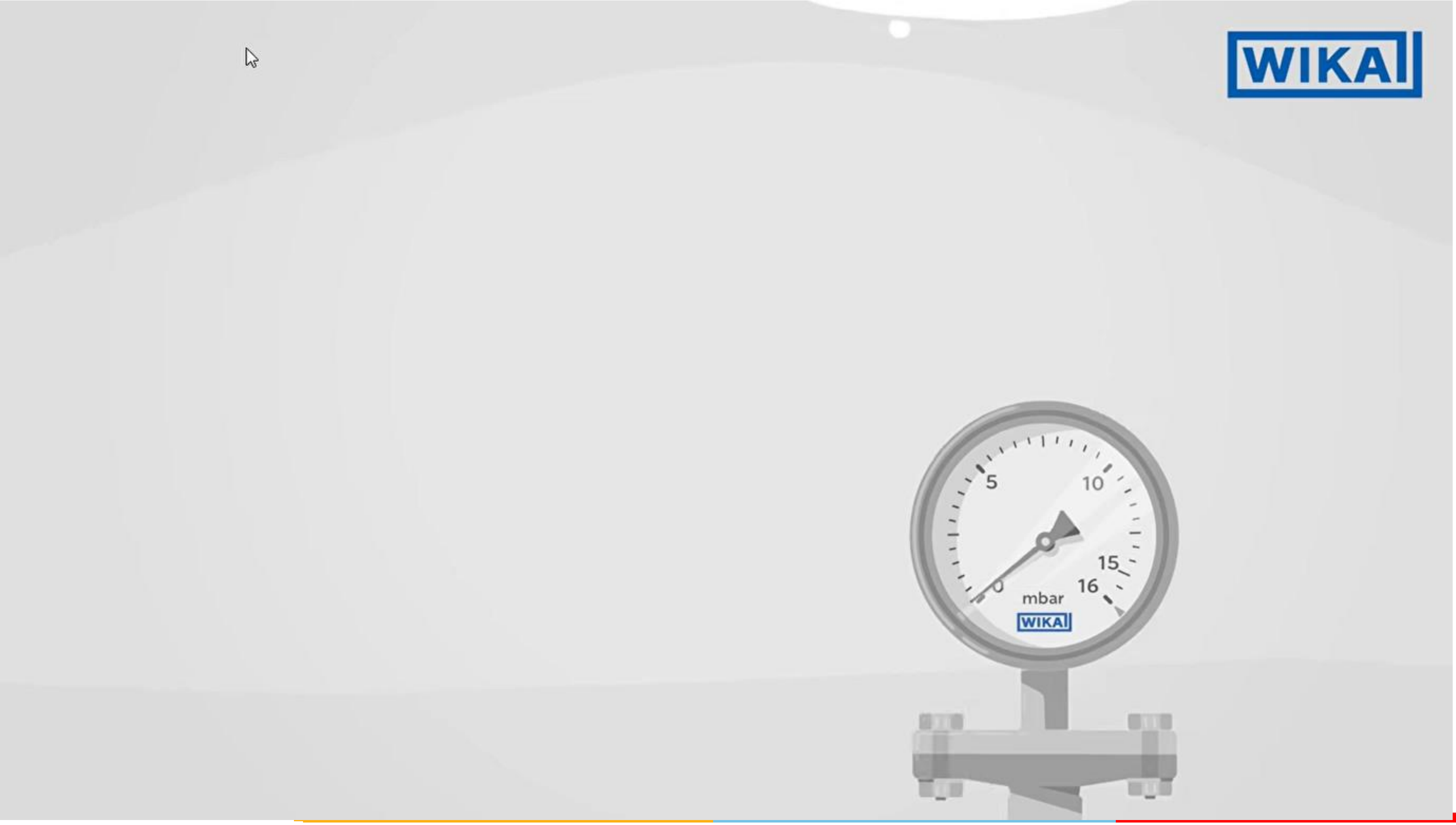
Mechanical Gauges

Bourdon tube pressure gauge (measuring spans 0-0.6 bar to 0-6000 bar)

https://www.youtube.com/watch?v=9avIVZP1Ka4

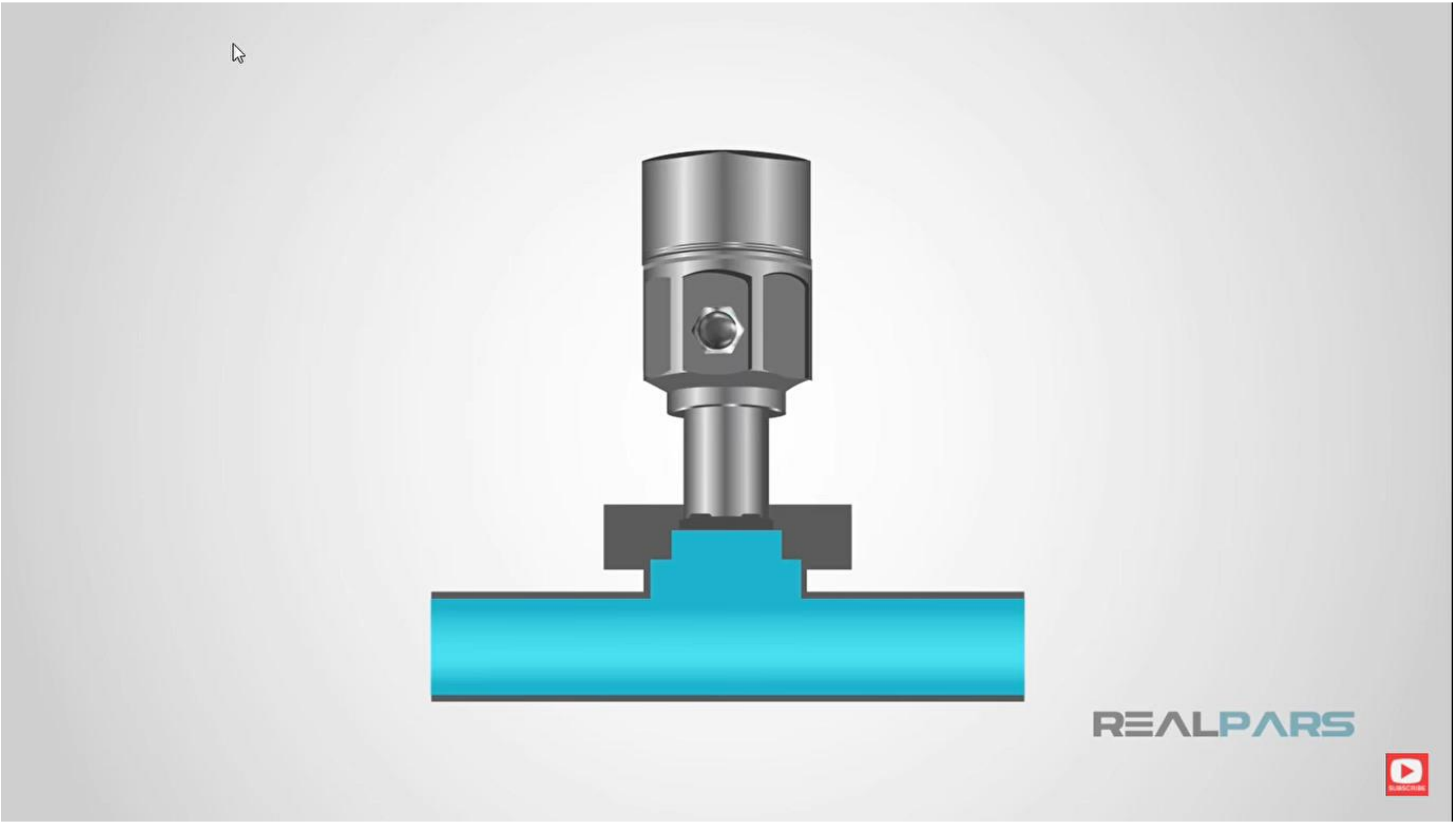
Mechanical Gauges

Diaphragm pressure gauge (measuring low pressures 0-0.016 bar)



https://www.youtube.com/watch?v=JnPSeEW-EeA&t=90s

Pressure Sensors



https://www.youtube.com/watch?v=iru8tRwS7Yc&t=135s

THANK YOU